

# Electrolysis of water and aqueous solutions (HL)

## Higher level (HL)

### Learning outcomes

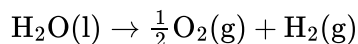
By the end of this section you should be able to:

- Deduce from standard electrode potentials, the products of the electrolysis of aqueous solutions.
- Demonstrate that electrolytic processes include the electrolysis of water and of aqueous solutions.
- Describe the effects of concentration and the nature of the electrode for the electrolysis of NaCl(aq) and CuSO<sub>4</sub>(aq).

As you learned in [section R3.2.13–14](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/14%20gibbs-free-energy-and-standard-electrode-potentials-hl-id-44784/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/14%20gibbs-free-energy-and-standard-electrode-potentials-hl-id-44784/>) electrochemical reactions can occur spontaneously or non-spontaneously. Non-spontaneous electrochemical reactions require a constant supply of energy to drive the reaction forward, in the form of an external power supply. Recall that it is also possible to oxidise and reduce the component ions in a molten salt through the use of an external power supply, despite the reaction being non-spontaneous (see [section R3.2.8](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electrolytic-cells-id-46491/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electrolytic-cells-id-46491/>)). In fact, electrolysis reactions can occur not only with molten salts but in aqueous solutions as well. How does the presence of water in the solution change the products of electrolysis? To answer this question, we will begin by looking at the electrolysis of water.

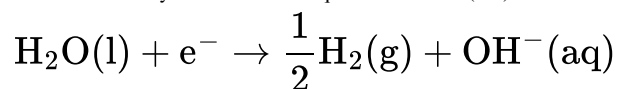
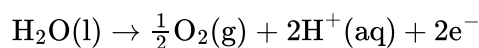
## Electrolysis of water: A non-spontaneous redox process

If it is possible for aqueous solutions to undergo electrolysis reactions, then it stands to reason that it should be possible for the water in the aqueous solution to be oxidised and reduced. This is referred to as the electrolysis of water, which occurs according to the overall (net) chemical reaction:



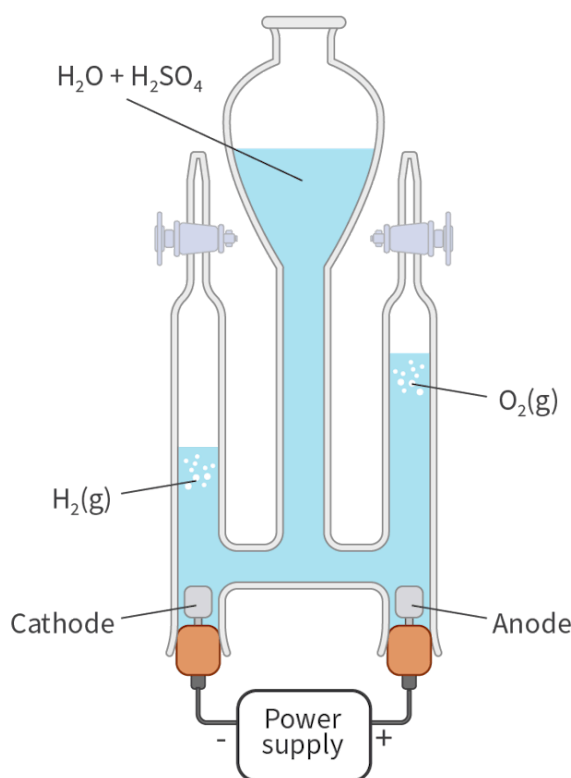
This net reaction can be further subdivided into the possible reactions taking place at the anode and the cathode.

**Reduction half-reaction (at the cathode):**


**Oxidation half-reaction (at the anode):**


All half-reactions and their standard electrode potentials are provided in [section 19 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/standard-reduction-potentials-at-29815-k-id-45121/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/standard-reduction-potentials-at-29815-k-id-45121/) of the DP chemistry data booklet.

**Figure 1** shows the apparatus used for the electrolysis of water, which is known as a Hofmann voltameter. The water to be electrolysed is mixed with dilute sulfuric acid,  $\text{H}_2\text{SO}_4(\text{aq})$ , which is necessary because pure water is a poor conductor of electricity. The addition of the dilute sulfuric acid (dilute sodium hydroxide solution,  $\text{NaOH}(\text{aq})$ , can also be used) allows the solution to conduct electricity due to the presence of mobile ions. So rather than pure water being electrolysed, it is actually a dilute solution sulfuric acid or sodium hydroxide. Based on the net equation for the reaction shown above, what is the ratio of  $\text{H}_2(\text{g})$  to  $\text{O}_2(\text{g})$  produced?



**Figure 1.** The electrolysis of water using a Hoffman voltameter. What is the ratio of hydrogen gas to oxygen gas produced in the reaction?

 More information for figure 1

The image is a diagram of a Hoffman voltameter used for the electrolysis of water. It illustrates the setup with a central reservoir containing water mixed with sulfuric acid, represented as  $\text{H}_2\text{O} + \text{H}_2\text{SO}_4$ . The central container is connected to two side tubes: one labeled with  $\text{H}_2(\text{g})$ , representing hydrogen gas, and the other with  $\text{O}_2(\text{g})$ , representing oxygen gas. Hydrogen gas is produced at the cathode, while oxygen gas is produced at the anode, both part of the power supply labeled at the bottom of the setup. The hydrogen is

collected on the left, and oxygen on the right, with labeled outlets for each gas. The diagram shows the collection of gases in the respective tubes, illustrating the 2:1 volume ratio of hydrogen to oxygen produced in the reaction.

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The addition of a universal indicator to the electrolyte shows interesting qualitative data, as the indicator turns from green to orange at the anode (due to the production of hydrogen ions) and from green to blue at the cathode (due to the evolution of hydroxide ions). Other qualitative and quantitative observations can be made by collecting the gas phase products at each electrode. From the net reaction for the electrolysis of water it is possible to see that hydrogen gas is produced at twice the rate of oxygen gas, as two moles of hydrogen gas are produced for every mole of oxygen gas. This is also reflected in the volumes of the gases produced in **Figure 1** - the ratio of hydrogen to oxygen is 2:1.

### Worked example 1

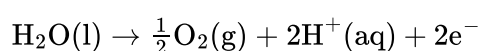
The electrolysis of water is conducted in a lab using a dilute sodium hydroxide solution, two graphite electrodes with attaching wires to a 9 V battery. When the reaction takes place, small bubbles start to form at both electrodes. Suggest two possible experimental techniques that could be used to determine the nature of the gases being evolved at each of the electrodes.

- Use a universal indicator, pH paper strips, or a pH probe to determine the change in pH at each electrode. This would allow you to determine the rate of formation of the hydroxide ions and hydronium ions at each electrode.
- Collection of each gas at either electrode using an eudiometer would be possible. It would be possible to measure the rate of formation of each gas evolved.
- Conducting a flaming splint test and a glowing splint test on each gas sample would allow you to identify the hydrogen and oxygen gas.

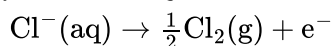
## The effect of concentration on the electrolysis of aqueous solutions

There is a special consideration required when an aqueous solution of sodium chloride,  $\text{NaCl(aq)}$ , is electrolysed. Unlike the electrolysis of molten sodium chloride – which has only one possible product at the anode – the presence of water in the aqueous solution somewhat complicates the situation. This is because, as discussed previously, water itself can undergo oxidation to produce oxygen gas,  $\text{O}_2(\text{g})$ .

The half equation for the oxidation of water is as follows:



And the half equation for the oxidation of chloride ions is:

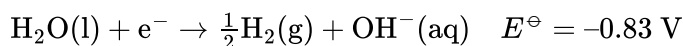
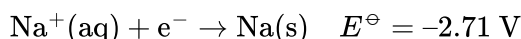


So which species will undergo oxidation at the anode and which gas will be produced? The answer to this question depends on the concentration of the solution.

- In dilute solutions of sodium chloride, the water will be oxidised leading to the production of oxygen gas at the anode.
- In concentrated solutions, the chloride ions will be preferentially oxidised leading to the formation of chlorine gas.

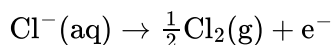
Changes in the concentration of the salt solution have a large impact on which species is oxidised during electrolysis. In dilute solutions, water molecules are oxidised to produce oxygen gas. Whereas, in concentrated solutions of sodium chloride, chloride ions are oxidised to produce chlorine gas. So to summarise, if the concentrated solution contains a halide ion ( $\text{Cl}^{-}$ ,  $\text{Br}^{-}$  or  $\text{I}^{-}$ ) then the halogen ( $\text{Cl}_2$ ,  $\text{Br}_2$  or  $\text{I}_2$ ) is given off at the anode. However, if the halide ions are present in a very low concentration, oxygen gas will be given off at the anode instead.

The two possible reactions at the cathode are the reduction of sodium ions or the reduction of water:

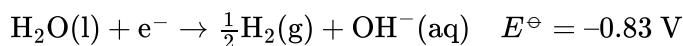


$\text{H}_2\text{O}$  is a stronger oxidising agent than  $\text{Na}^{+}$  ions ( $\text{H}_2\text{O}$  has a more positive  $E^{\ominus}$  value), meaning that it is more easily reduced. Therefore, at the cathode, water undergoes reduction to form hydrogen gas. The electrolysis of concentrated sodium chloride solution (also known as brine) is the basis of a major chemical industry noted for its efficient use of all of the products. The electrolysis of brine produces hydrogen and chlorine gases, but the product of greatest economic importance is the alkali, sodium hydroxide. Note that sodium hydroxide is not produced at either the cathode or the anode of the cell, instead, it exists as the solution formed in the reaction.

In general, metal ions with a less positive  $E^{\ominus}$  value than water will not be reduced at the cathode when part of an aqueous solution. And metal ions with a more positive  $E^{\ominus}$  value than water will be reduced instead of water. Take, for example, a concentrated aqueous solution of copper(II) chloride,  $\text{CuCl}_2(\text{aq})$ . Because of the high concentration of chloride ions, the product at the anode is chlorine gas.



At the cathode, the two possible half equations are the reduction of water:



or the reduction of copper(II) ions:



The  $E^{\ominus}$  value for the reduction of copper(II) ions is more positive than that for the reduction of water. For this reason, copper(II) ions are reduced to form copper metal.

Use **Interactive 1** to investigate the gases given off at the anode and the cathode.

Chlorine gas

Hydrogen gas

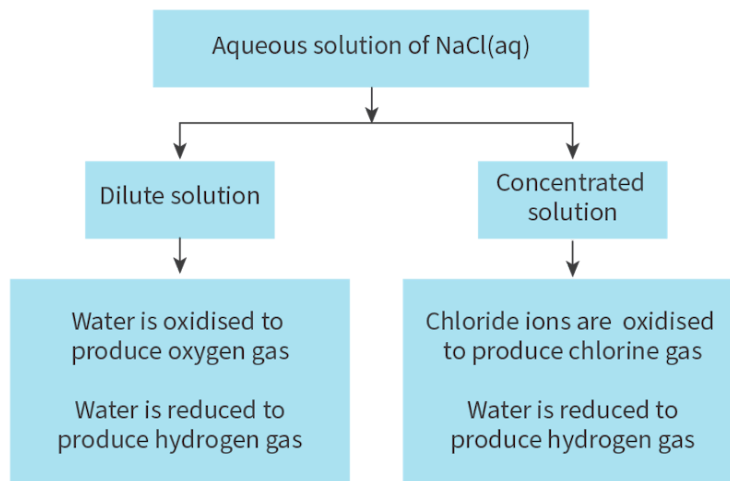
Oxygen gas

Copper solid

☒ Check

### Interactive 1. Electrolysis of aqueous solutions at different concentrations.

**Figure 2** shows a concept diagram of the electrolysis of aqueous solutions at different concentrations.



**Figure 2.** A concept diagram of the electrolysis of aqueous solutions.

 More information for figure 2

The diagram illustrates the electrolysis of an aqueous NaCl (sodium chloride) solution, divided into two paths based on concentration. At the top, a blue box labeled "Aqueous solution of NaCl(aq)" is shown. Two arrows branch downward from this box.

On the left, the path is labeled "Dilute solution." Below, a box contains the text: "Water is oxidised to produce oxygen gas. Water is reduced to produce hydrogen gas."

On the right, the path is labeled "Concentrated solution." Below, a box contains the text: "Chloride ions are oxidised to produce chlorine gas. Water is reduced to produce hydrogen gas."

Each process is connected by arrows, indicating the flow of chemical reactions based on the concentration of the solution.

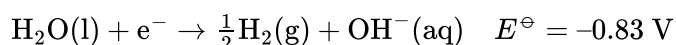
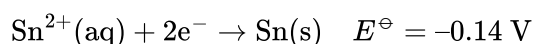
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## Worked example 2

Tin(II) bromide is highly soluble in water. Predict the products formed via the electrolysis of a concentrated solution of  $\text{SnBr}_2(\text{aq})$  at SATP.

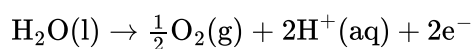
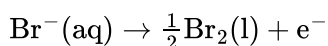
The possible half-reactions occurring at the cathode and anode must be assessed based on their standard electrode potentials to see which species is preferentially oxidised and which is preferentially reduced. In addition, because the solution contains aqueous halide ions,  $\text{Br}^-(\text{aq})$ , the concentration must also be considered.

Cathode (reduction):



Tin ions have a more positive (or less negative)  $E^\ominus$  value so will be preferentially reduced to form solid tin at the cathode.

Anode (oxidation):



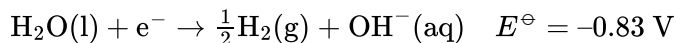
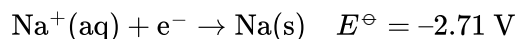
In a concentrated solution, bromide ions will be preferentially oxidised to form liquid bromine at the anode.

## Worked example 3

A concentrated sodium iodide solution can undergo electrolysis at SATP to produce hydrogen gas and iodine. Determine the half-equations and products at each electrode.

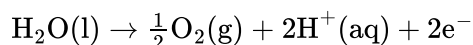
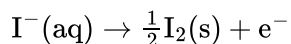
The possible half-reactions occurring at the cathode and anode must be assessed based on their standard electrode potentials to see which species is preferentially oxidised and which is preferentially reduced.

Cathode (reduction):



Hydrogen gas will be evolved at the cathode as the hydrogen is preferentially reduced over the sodium ions at the cathode.

Anode (oxidation):



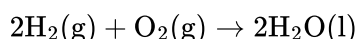
Because of their high concentration, iodide ions will be preferentially oxidised at the anode to form iodine.

### Making connections

You previously learned that the use of hydrogen fuel cells, which are voltaic cells, allows for the continuous production of an electrical current (see section R1.3.5 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/fuel-cells-id-44783/>)). This type of fuel cell produces an electrical current simply by subdividing the spontaneous redox reaction between oxygen and hydrogen to produce water and harnessing the flow of electrons between the two half-cells in the wire.

It is interesting to note that the hydrolysis of water and the reaction taking place in a hydrogen fuel cell are the reverse of the same reaction.

Where one process



is spontaneous and releases energy, the hydrolysis of water is a non-spontaneous process and requires the input of energy.

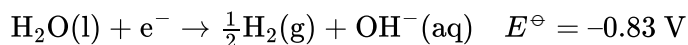
## Types of electrodes and electrolytes on the electrolysis of salt solutions

Up until this point you have looked extensively at the electrolysis of aqueous salt solutions consisting of a metal cation and a halogen anion. As you have seen salts form not only from metal cations and halogen anions, but a variety of polyatomic species (see section R3.1.12 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ph-of-salt-solutions-hl-id-46060/>)). It is possible for these types of salt solutions to undergo electrolysis, but how they undergo electrolysis changes somewhat depending on the type of electrode used to electrolyse the solution.

Take for example the electrolysis of copper(II) sulfate using an inert electrode like platinum. Platinum is able to carry a charge into the electrolyte but not participate in the chemical reaction. Graphite can also be used as an inert electrode as it is relatively unreactive, has a high melting point, and is a good electrical conductor

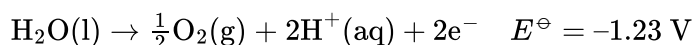
due to the delocalised electrons in its molecular structure that provides for good electron mobility. It is essential to use electrodes which are highly conductive, have high melting points, and yet are not preferentially oxidised or reduced over the species in solution.

At the cathode the possible reactions are:



Copper(II) ions will be preferentially reduced and copper solid will be produced at the cathode.

At the anode the only possible reaction is:



The sulfur atom in  $\text{SO}_4^{2-}$  has an oxidation state of +6, which is its highest oxidation state, and therefore cannot be further oxidised. This is true for many other polyatomic oxyanions like chlorate, nitrate, and phosphate to name a few.

### Worked example 4

The electrolysis of a copper(II) sulfate solution takes place using two graphite electrodes. What would be possible signs that a chemical reaction is taking place once the electrolysis starts to occur?

Use a universal indicator, pH paper strips, or a pH probe to determine the change in pH would show a decrease in the pH of the electrolyte as hydronium ions are produced.

- Production of a goldish orange colour on the cathode, as copper solid is deposited.
- Production of oxygen gas bubbles at the anode. If collected the gas could be identified using a glowing splint test.
- The intense blue colour of the copper(II) sulfate solution would decrease over time due to the removal of the copper(II) ions via reduction.

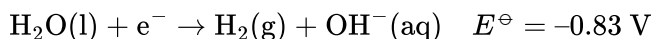
### Worked example 5

An aqueous solution of concentrated copper(II) sulfate undergoes electrolysis using inert carbon electrodes.

State the half-reactions that occur at the cathode and the anode, and the overall balanced chemical reaction for the electrolysis of concentrated copper(II) sulfate.

At the cathode, the two following reduction reactions could be taking place:



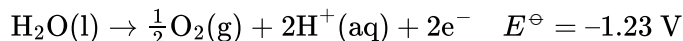


As the copper(II) ions have a more positive standard electrode, they will be preferentially reduced compared to the water.

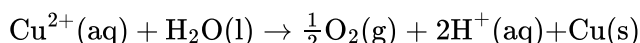
At the anode, two possible oxidation half-reactions should be able to occur, that of water or the oxidation of the sulfate ion.

Note: The sulfur atom in  $\text{SO}_4^{2-}$  has an oxidation state of +6, which is its highest oxidation state, and therefore cannot be further oxidised.

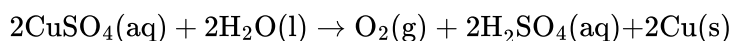
This means that at the anode, water is being preferentially oxidised:



The overall net-ionic equation is therefore the sum of these two half reactions:



Adding in the spectator sulfate ions and doubling the coefficients produces the reaction:



If a reactive electrode like copper is used instead of an inert electrode then the solid copper electrode will be preferentially oxidised at the anode and the electrode will produce copper(II) ions in solution instead of generating oxygen gas.

This means that the pH of the electrolyte would remain constant, as would the concentration of copper(II) ions in solution. Copper would still be plated onto the cathode and the copper anode would be slowly eroded.

### Aspect: Experiments

An extremely important aspect of scientific research often has to do with improving the protocols of other scientists. Developing new techniques allows us to improve the yield of chemical reactions and is an essential part to most areas of chemical research.

For example, the direct-current electrolysis of concentrated potassium hydroxide solutions often needs to occur at high temperatures (up to 400 °C) and under extremely high pressures. From your studies of chemical kinetics, it is easy to see the advantages of running an electrolysis reaction at high temperatures and pressures, including increasing the conductivity of the electrolyte and increasing the rates of electrochemical reactions at the electrode surfaces.

Researchers often need to design experiments in order to meet the constraints of the chemical reactions they are studying. In this example, Jason C. Ganley of Howard University was able to better understand the type of anode materials that would work best under these conditions,

including nickel, lithiated nickel, cobalt-plated nickel and other metal alloys in this article [↗](#)

(<https://www.sciencedirect.com/science/article/abs/pii/S0360319909003577?via%3Dihub>).

## Activity

- **IB learner profile attribute:**
  - Communicator
  - Reflective
- **Approaches to learning:** Communication skills — Clearly communicating complex ideas in response to open-ended questions; Using terminology, symbols and communication conventions consistently and correctly
- **Time required to complete activity:** 20—30 minutes
- **Activity type:** Individual/group activity

In this activity you will use your understanding of electrochemical reactions and standard cell potentials to reflect on the conditions necessary to produce an electrolysis reaction of an aqueous solution.

### Your task

You will analyse the electrolysis reaction and describe the properties and components of your chosen system. The ability to communicate complex concepts correctly and clearly to others is a vital skill. As well, in preparing your ideas and concepts for explanation you often develop a better understanding of the processes.

**Table 1** contains the half-reactions and associated standard electrode potentials for a variety of different species that can combine to produce an ionic salt. You should select a solution that when electrolysed would produce solid metal at the cathode and oxygen gas at the anode.

**Step 1:** Select from **Table 1** two species that would form an appropriate ionic salt.

**Step 2:** Sketch a diagram of the electrolytic cell for your chosen salt in aqueous solution.

- Include the type of electrode required (inert or reactive).
- The half-reactions occurring at each electrode.
- The approximate molecular concentration of the aqueous solution (very dilute, dilute or concentrated).
- The products being produced at each electrode.
- The overall balanced reaction for your chosen electrolysis process.

**Table 1.** Half-reactions and associated standard elec

| Oxidised species    | Reduced species |
|---------------------|-----------------|
| $X_2(g) + 2e^-$     | $2X^-(aq)$      |
| $Y^{2+}(aq) + 2e^-$ | $Y(s)$          |
| $W^{3+}(aq) + 3e^-$ | $W(s)$          |
| $Z_2(q) + 2e^-$     | $2Z^-(aq)$      |

Note: The standard electrode potentials and half-reactions for the oxidation and reduction of water are found in the DP chemistry data booklet.

### Extension

Share your sketch with the other members of your group and compare and contrast the half-reactions each of your group members chose. Discuss any similarities and differences between your chosen systems and look for any inconsistencies or areas where other students might need greater clarity.